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Artificial Intelligence (AI) Training Program

Course Overview

This intensive 2-day AI training program is designed to provide college students with a strong foundational and practical understanding of Artificial Intelligence. The course covers core AI concepts, Machine Learning fundamentals, Generative AI, and real-world applications with hands-on exposure using industry tools.

Participants will gain both conceptual clarity and practical skills to understand how AI systems are built and applied across industries.

⏱️ Duration

2 Days – 16 Hours

Course Objectives

By the end of this training, participants will be able to:

- Understand core concepts of Artificial Intelligence and its evolution
- Differentiate between AI, Machine Learning, and Deep Learning
- Build basic Machine Learning models
- Understand how Generative AI tools work
- Apply AI concepts to real-world scenarios
- Gain awareness of AI career paths and industry use cases

Target Audience

- College Students (All Streams – Engineering, Science, Commerce)
- Beginners interested in AI & Machine Learning

- Students preparing for AI-related careers

Prerequisites

- Basic computer knowledge
- Basic mathematics (optional but helpful)
- No prior coding experience required

Course Outline

Module 1: Introduction to Artificial Intelligence

- What is Artificial Intelligence?
- History and Evolution of AI
- AI vs Machine Learning vs Deep Learning
- Types of AI (Narrow AI, General AI, Super AI)
- Real-world applications of AI (Healthcare, Finance, E-commerce, Education)

Module 2: Foundations of Machine Learning

- What is Machine Learning?
- Types of Machine Learning:
  - Supervised Learning
  - Unsupervised Learning
  - Reinforcement Learning
- Key Concepts: Features, Labels, Training Data, Models
- Overfitting vs Underfitting
- Introduction to Model Evaluation

Module 3: Data Handling & Preprocessing

- Importance of Data in AI
- Types of Data (Structured vs Unstructured)
- Data Cleaning Techniques
- Handling Missing Data
- Feature Engineering Basics
- Introduction to datasets

Hands-on: Working with sample datasets

Module 4: Introduction to Python for AI

- Basics of Python (Variables, Data Types, Functions)
- Introduction to Libraries:
  - NumPy
  - Pandas
  - Matplotlib
- Data manipulation and visualization

Hands-on: Simple Python programs for data handling

Module 5: Core Machine Learning Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- K-Nearest Neighbors (KNN)
- Clustering (K-Means)

Hands-on:

- Build a basic prediction model
- Visualization of results

Module 6: Introduction to Deep Learning

- What is Deep Learning?

- Neural Networks Basics
- How AI mimics the human brain
- Introduction to TensorFlow / PyTorch
- Use cases (Image recognition, speech processing)

Module 7: Generative AI & Modern AI Tools

- What is Generative AI?
- Large Language Models (LLMs)
- Tools & Platforms:
  - ChatGPT
  - AI Image Generators
  - AI Coding Assistants
- Prompt Engineering Basics

Hands-on:

- Writing effective prompts
- Using AI tools for productivity

Module 8: AI Project & Case Study

- Mini Project (Guided):
  - Example: Student Performance Prediction / Spam Detection
- Understanding real-world AI workflows
- Model training and basic evaluation

Module 9: AI Ethics & Responsible AI

- Bias in AI
- Ethical considerations
- Data privacy & security
- Responsible AI usage

Module 10: AI Careers & Industry Trends

- Career paths in AI:
    - Data Scientist
    - Machine Learning Engineer
    - AI Engineer
- Required skills and certifications
- Industry demand and future trends

Training Methodology

- Instructor-led interactive sessions
- Hands-on labs and exercises
- Real-world case studies
- Group discussions and activities
- Mini project implementation

Tools & Technologies Covered

- Python
- Jupyter Notebook / Google Colab
- NumPy, Pandas, Matplotlib
- Intro to TensorFlow / PyTorch
- Generative AI tools

Learning Outcomes

After completing this program, participants will:

- Understand end-to-end AI concepts
- Build simple machine learning models
- Work with data and basic AI tools
- Use Generative AI effectively
- Be ready to explore advanced AI learning paths

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Certification

Participants will receive a **myTectra Certificate of Completion in Artificial Intelligence Training**